

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: *Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization.(BL3,BL4,BL5)*

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25 Total Marks: 50
Annexure A

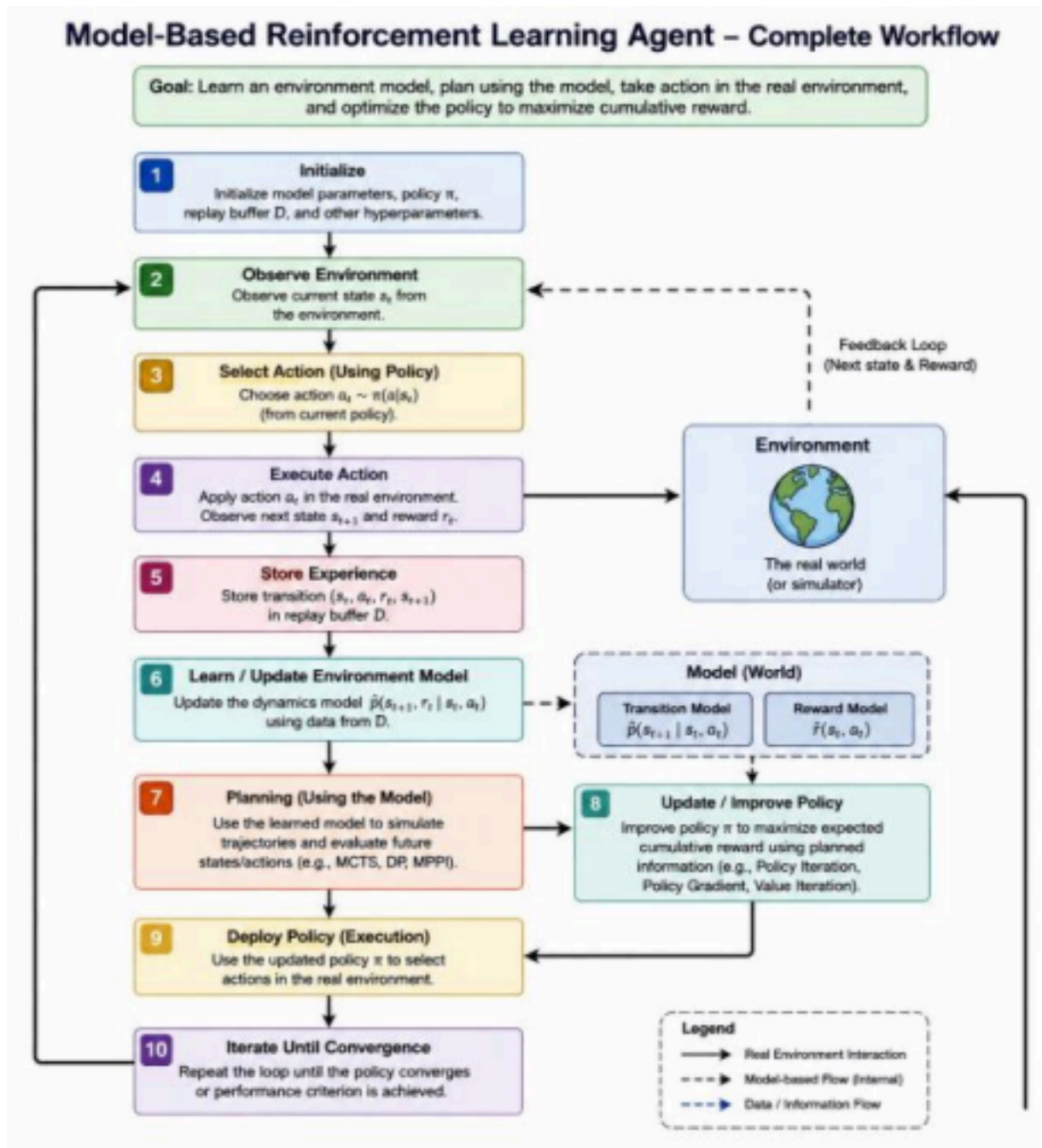
Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Model-Based RL Approach

1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

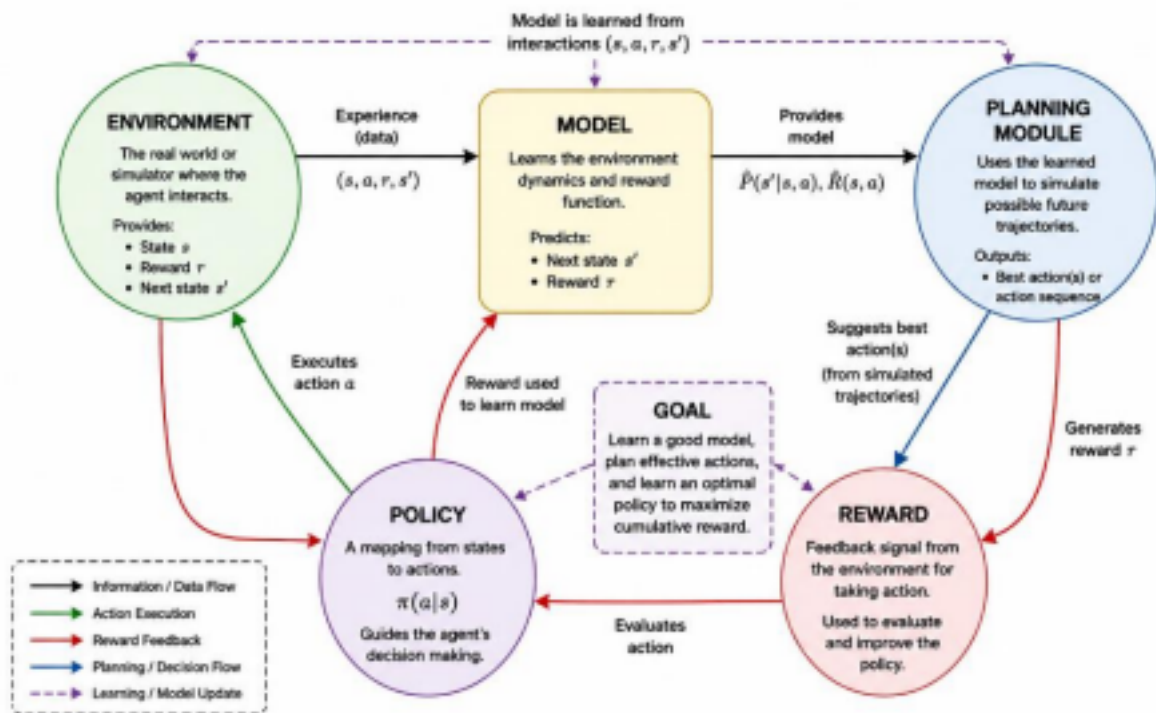
(5 Marks)



2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.

(5 Marks)

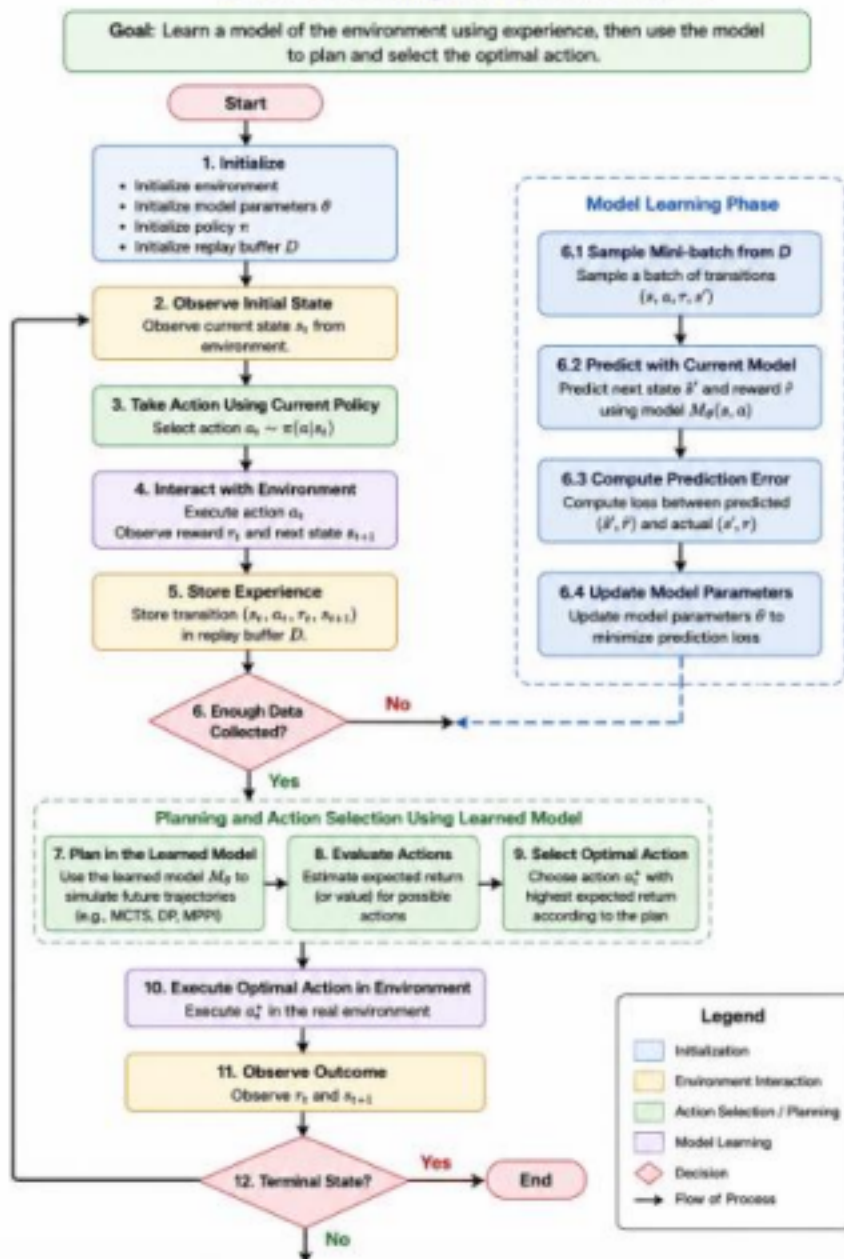
Concept Map: Components of Model-Based Reinforcement Learning



3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.

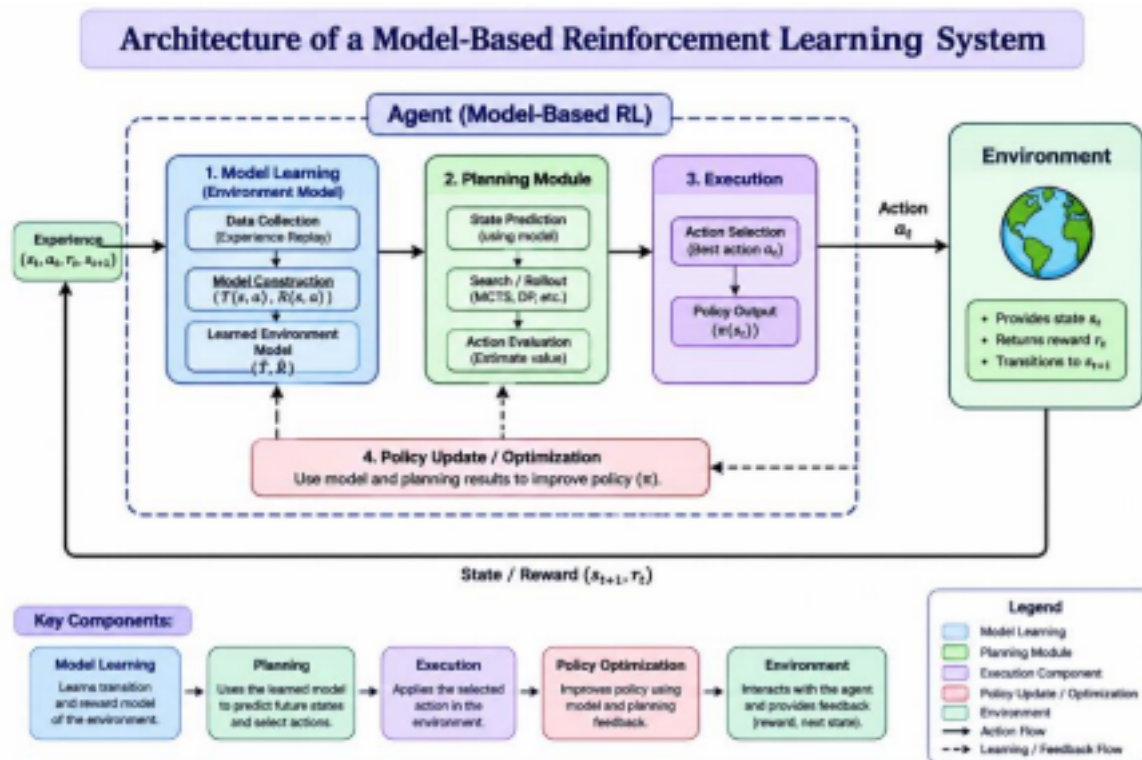
(5 Marks)

Flowchart: How an RL Agent Learns an Environment Model Before Selecting an Optimal Action



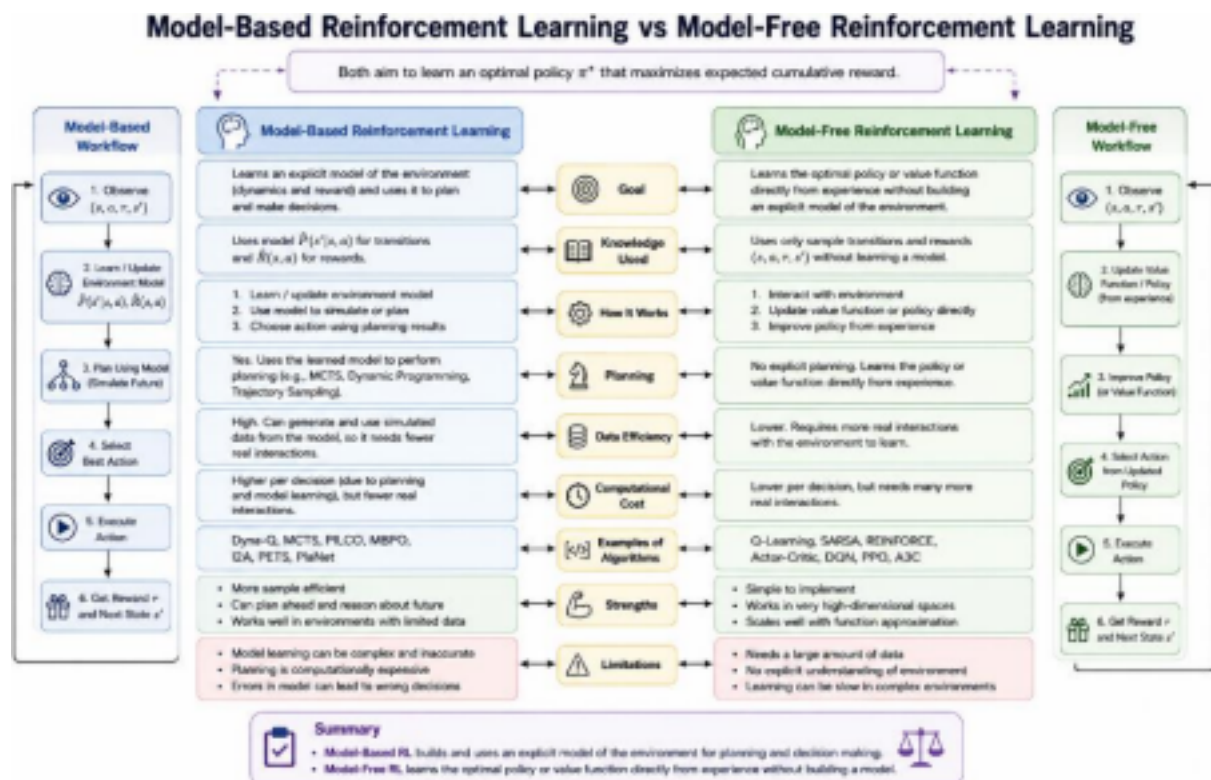
4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.

(5 Marks)



5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

(5 Marks)



Code of Conduct:

*I M.Meghana-192472351 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M.Meghana

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness